

Turbo-generator Set CB 24 GT4

Complete Manual

Order 7.318.042

Folder 1: Operating Manual with drawings

Folder 2: Installation requirements, coupling manual and accessory documentation

Operating Manual

Turbo-generator set CB 24 GT4

Back-pressure steam turbine with gearbox and generator

Order 7.318.042

Edition 04.2019

Read this manual before working on the turbo-generator set. Keep it at the machine.

Part A Operating Instructions

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Part B Drawings and diagrams

Arrangement drawing
Oil system diagram
Steam and drain diagram
Instrument list

Scope of documentation

The documentation of the turbo-generator set consists of this operating manual (Part A), the drawings and diagrams (Part B) and the site installation requirements for the turbine.

The coupling between gearbox and generator is described in the separate operating manual of the coupling manufacturer.

Notes on this manual

Safety notes are marked with the signal words danger, warning and caution. Danger marks an imminent hazard that leads to death or serious injury; warning marks a possible hazard; caution marks a hazard to the machine.

Values given in this manual apply to the machine as delivered. Values in the instrument list of Part B take precedence where they differ.

1 General notes

1.1 Preface

This operating manual describes the safe operation and maintenance of the turbo-generator set. This steam turbine was designed and built according to the engineering practice and the safety rules that applied at the time of delivery.

Operating personnel must be trained on the machine and must have read this manual, especially the chapter on safety, before they operate the set.

1.2 Intended use

The turbo-generator set converts the energy of process steam into electrical energy. The exhaust steam is used in the process steam header of the plant.

Any other use, operation outside the rated data, or modification without written approval of the manufacturer is not intended use.

1.3 Rated data

Live steam pressure, live steam temperature, back pressure, speed and electrical output are given on the rating plate and in the data sheet of Part B.

The turbine runs at its rated speed and drives the generator through a parallel-shaft gearbox.

1.4 Warranty and liability

The warranty expires when the machine is operated outside the rated data, when non-original spare parts are used, or when the maintenance schedule is not followed.

Keep the operating log up to date; it is required for warranty claims.

1.5 Personnel qualification

Operation, maintenance and repair are reserved for qualified personnel who are authorized by the plant owner.

Work on the electrical equipment is reserved for qualified electricians.

1.6 Documentation of changes

Record every change of settings, every repair and every replacement of parts in the operating log together with the date and the name of the person responsible.

1.7 Oil vapour extraction

The oil tank is kept under a slight vacuum by the oil vapour extractor. The extracted air passes through the oil mist separator, which returns the separated oil to the tank.

A blocked oil mist separator raises the pressure in the bearing housings and causes oil leakage at the shaft seals.

1.8 Units and abbreviations

Pressures are given as gauge pressure unless stated otherwise. Temperatures are given in degrees Celsius.

Abbreviations used in this manual are listed in the instrument list of Part B.

1.9 Environmental protection

Used oil, filter elements and oil-soaked cleaning material must be disposed of according to local regulations. Do not let oil or cooling water with additives enter the drainage system.

1.10 Contact

For questions about operation and spare parts contact the service department of the manufacturer and quote the order number of the machine.

Service: service@rkt-turbines.example

2 Safety

2.1 General safety notes

Observe the safety notes in this chapter and in the individual chapters of this manual. Protective devices must never be removed or bridged while the set is in operation.

2.2 Hazards from steam and hot surfaces

Danger: escaping steam causes severe burns. Do not open flanges, drains or instrument connections while the pipes are under pressure.

Warning: casing and valves remain hot for many hours after shutdown.

2.3 Hazards from rotating parts

Warning: never reach into the coupling area while the shaft is turning. Fit the coupling guards before start-up.

2.3.1 Noise

The sound level near the running set exceeds the limit for unprotected hearing. Wear hearing protection in the turbine area.

2.3.2 Fire protection

Oil leaking onto hot surfaces can ignite. Repair leaks immediately and replace oil-soaked insulation. Keep fire extinguishers ready near the oil system.

2.3.3 Lifting and transport

Use only the lifting points shown in the arrangement drawing. Never stand under suspended loads.

2.4 Notes on safety for operation

The turbine protection shuts the turbine down automatically by closing the quick-closing valve when one of the following limits is exceeded. Each trip is preceded by an alarm at a lower limit so that the operator can react in time.

- Overspeed of the turbine rotor.
- Live steam pressure too high or too low.
- Live steam temperature too high.
- Bearing temperature too high.
- Lube oil pressure too low.
- Vibrations too high.
- Axial position of the rotor out of limits.

Warning: after a trip the turbine may be restarted only when the cause for the trouble has been found and eliminated. Record every trip and alarm in the operating log. Repeated restarts without eliminating the cause can damage the bearings and the blading.

2.5 Testing the protective devices

Test the overspeed trip and the quick-closing valve at the intervals of the maintenance schedule and after every repair of the governor. Record the trip speed in the operating log.

2.6 Emergency stop

The emergency stop buttons in the control room and at the machine close the quick-closing valve and open the generator breaker. Use the emergency stop only in case of danger.

2.7 Personal protective equipment

Wear safety shoes, hearing protection, gloves and eye protection when working at the machine. Wear heat-resistant gloves when working on hot parts.

3 Description of the turbo-generator set

3.1 Overview

The set consists of the back-pressure steam turbine, the gearbox, the generator, the oil system and the control system, mounted on a common base frame.

3.2 Turbine casing

The casing is split horizontally. The upper half can be removed for inspection of the rotor and the blading without disconnecting the steam pipes.

3.3 Turbine rotor

The rotor carries a velocity-compounded control stage and several reaction stages. It runs in two tilting-pad journal bearings and one tilting-pad thrust bearing.

3.4 Shaft seals

Labyrinth seals at both shaft ends limit the leakage of steam to atmosphere. Leak-off steam is led to the gland steam condenser.

3.5 Quick-closing valve and control valves

The quick-closing valve shuts off the live steam in case of a trip. The control valves are moved by a hydraulic servomotor and set the steam flow according to the governor signal.

3.6 Gearbox

The parallel-shaft gearbox reduces the turbine speed to the generator speed. Its bearings are lubricated from the common oil system.

3.7 Generator

The generator is a brushless synchronous machine with air cooling through a water-cooled heat exchanger. Stator winding temperatures are monitored by resistance thermometers.

3.8 Oil system

The main oil pump is driven by the gearbox. An auxiliary oil pump supplies oil during start-up and shutdown, and an emergency oil pump supplies the bearings during coast-down after a power failure.

3.9 Governor

The electronic governor controls speed during start-up and back pressure or load during operation. It also contains the electronic overspeed protection.

3.10 Monitoring

Bearing temperatures, shaft vibration, lube oil pressure, axial position, speed and steam values are shown on the operator panel and passed to the plant control system.

4 Transport and installation

4.1 Transport

The set is delivered on its base frame. Lift it only at the marked lifting points and keep it level during transport.

4.2 Storage

If the set is stored before installation, keep it dry and protected against dust. Turn the rotor by a quarter of a turn every month to protect the bearings.

4.3 Installation

Install the set according to the site installation requirements for the turbine. Check the foundation and the anchor bolts before setting the base frame.

4.4 Alignment

Check the alignment of turbine, gearbox and generator after setting the base frame and again after grouting and after connecting the pipes.

4.5 Connecting the piping

Connect the steam, drain, oil and cooling water pipes free of stress. Remove all transport covers and blank flanges before connecting.

4.6 Filling the oil system

Fill the oil tank with the specified oil through a fine filter. Flush the oil system until the filters remain clean.

4.7 Electrical connection

Connect the generator, the auxiliary drives and the control cabinet according to the circuit diagrams. Check the direction of rotation of the pump motors.

4.8 Insulation

Insulate the casing and the steam pipes after the pressure test. Keep the flange joints accessible.

4.9 Preservation before commissioning

If commissioning is delayed, keep the preservation of the turbine and the oil system in place and check it regularly.

4.10 Handover for commissioning

Hand over the set to the commissioning team when the installation checklist is completed and signed.

5 Commissioning

5.1 Preconditions

Before the first start the oil system must be flushed, the steam pipes blown out, the protective devices tested and the alignment recorded.

5.2 Checking the oil system

Start the auxiliary oil pump and check oil pressure, oil temperature and the oil flow at every bearing. Check the oil system for leaks.

5.3 Checking the protective devices

Test the quick-closing valve, the overspeed trip and every protection signal before steam is admitted to the turbine.

5.4 Warming up

Open the drains and warm up the live steam pipe slowly. Admit steam to the turbine only when the pipe is drained and warm.

5.5 First run-up

Run the turbine up to low speed and listen for rubbing noise. Hold the speed and check bearing temperatures and vibration before increasing speed further.

5.6 Overspeed test

Raise the speed slowly until the overspeed trip closes the quick-closing valve. Record the trip speed and compare it with the setting in the instrument list.

5.7 Synchronizing

Synchronize the generator with the grid using the automatic synchronizer. Then take over load in small steps.

5.8 Load operation

Increase the load slowly while watching bearing temperatures, vibration and axial position. Switch the governor to back-pressure control when the process header can take the steam.

5.9 Commissioning report

Record all set values, test results and measured values in the commissioning report.

5.10 Trial run

The trial run lasts until the set has run at full load under stable conditions for the time agreed in the contract.

6 Operation

6.1 Start-up from cold condition

Start the auxiliary oil pump, open the drains and warm up the steam pipes. Run up the turbine following the start-up curve in Part B.

6.2 Start-up from warm condition

After a short standstill the turbine can be started with a shorter warm-up time. Follow the start-up curve for warm condition.

6.3 Monitoring during operation

Check bearing temperatures, vibration, lube oil pressure, oil temperature and steam values on every shift and record them in the operating log.

6.4 Load changes

Change the load slowly. Large and fast load changes cause thermal stress in the casing and the rotor.

6.5 Operation in island mode

In island mode the governor controls speed and frequency. Observe the limits for load steps given in Part B.

6.6 Normal shutdown

Reduce the load, open the generator breaker and close the quick-closing valve. The auxiliary oil pump starts automatically when the speed falls.

6.7 Turning after shutdown

Keep the auxiliary oil pump running until the bearing temperatures have fallen to the value given in the instrument list.

6.8 Operating log

Record start-ups, shutdowns, alarms, trips and all abnormal observations in the operating log.

6.9 Rounds

On each round check the machine for leaks, unusual noise and unusual smell, and check the level in the oil tank.

6.10 Operation with reduced back pressure

Operation below the minimum back pressure is not permitted because it overloads the last stage blading.

7 Maintenance

7.1 Maintenance schedule

Carry out maintenance at the intervals of the maintenance schedule in Part B. Shorten the intervals under severe operating conditions.

7.2 Daily checks

Check oil level, oil temperature, oil pressure, leaks and the readings of the monitoring system.

7.3 Weekly checks

Test the auxiliary oil pump and the emergency oil pump by a short test run. Check the oil filters and switch over to the clean filter when the differential pressure is high.

7.4 Monthly checks

Take an oil sample and have it analysed for water content, particles and ageing. Check the free movement of the quick-closing valve by a partial stroke test.

7.5 Oil change

Change the oil when the oil analysis shows that the limit values are exceeded. Clean the oil tank before refilling.

7.6 Oil filters

Replace the filter elements when the differential pressure alarm occurs. Vent the filter housing after replacing the element.

7.7 Oil mist separator

Check the oil mist separator on the oil tank vent for oil return and pressure drop. Replace the separator element when the tank pressure rises or oil escapes at the vent.

A clogged oil mist separator causes oil leakage at the bearing housings. Check the drain line of the separator for free flow.

7.8 Coolers

Clean the oil cooler and the generator air cooler on the water side when the cooling performance falls.

7.9 Steam strainer

Inspect the steam strainer at every major inspection and remove deposits.

7.10 Governor and servomotor

Check the servomotor for leaks and free movement. Leaking oil seals at the servomotor must be replaced.

7.11 Bearings

Inspect the bearing pads at the major inspection. Measure the bearing clearance and replace pads with worn white metal.

7.12 Blading

Inspect the blading for deposits, erosion and damage when the casing is opened.

7.13 Coupling

Check the alignment and the coupling at the intervals given by the coupling manufacturer.

7.14 Generator

Check the insulation resistance of the stator winding and clean the air cooler as described in the generator documentation.

7.15 Major inspection

At the major inspection open the turbine casing, inspect rotor, blading, seals and bearings, and replace worn parts.

8 Trouble, causes and their elimination

This chapter tells you what to check first when the turbine trips, alarms or runs abnormally, and what must be done before restarting. The turbine protection trips the turbine on overspeed, high bearing temperature, low lube oil pressure, high vibration and wrong axial position. An unplanned shutdown always has a cause that must be found and eliminated before the turbo-generator set is brought back online.

Before restarting after a trip or an alarm shutdown: read the alarm list, check bearing temperatures and vibration trends before the trip, check the lube oil pressure and the oil level, and inspect the machine for leaks and unusual noise. Recover the set only after the cause has been eliminated and recorded in the operating log.

Trouble	Cause	Elimination
Lube oil pressure too low or dropped	Oil filter clogged; oil level too low; pump worn	Switch filter; top up oil; check the pump before restarting
Bearing temperature too high	Oil too hot; oil flow too low; bearing damaged	Clean the oil cooler; check oil flow; inspect the bearing pads
Vibrations too high	Misalignment; unbalance; bearing damage	Check alignment and coupling; balance the rotor; inspect bearings

8.1 Trouble table (continued)

Trouble	Cause	Elimination
Turbine trips on overspeed	Governor fault; control valve sticking	Check the governor and the valve stems before restarting
Leaking oil seal at the servomotor	Seal worn	Replace the seal
Speed hunts	Governor settings; air in the control oil	Adjust the governor; vent the control oil system
Leaking steam valve	Valve seat damaged	Grind in or replace the valve seat
Water in the oil	Leaking oil cooler; gland steam leakage	Check the cooler; check the gland steam system; change the oil
Unsmooth running	Rubbing; deposits on the blading	Inspect the rotor and the blading

8.2 Restart after a trip

After the cause of the trip has been eliminated, reset the protection, start the auxiliary oil pump and restart the turbine following the start-up procedure for warm condition. Watch bearing temperature and vibration closely during run-up.

If the turbine trips again for the same reason, do not restart it. Contact the service department of the manufacturer.

9 Shutdown and conservation

9.1 Short standstill

For a standstill of a few days keep the auxiliary oil pump running for some hours every day and keep the drains open.

9.2 Long standstill

For a longer standstill dry the turbine with warm air and fill the oil system with preservation oil.

9.3 Conservation of the steam path

Keep the steam path dry. Close the quick-closing valve and the exhaust valve and open the casing drains.

9.4 Conservation of the oil system

Circulate the oil regularly and check the oil for water content during the standstill.

9.5 Conservation of the generator

Keep the generator winding dry by switching on the anti-condensation heater.

9.6 Recommissioning after conservation

Remove the preservation, check the oil and carry out the commissioning steps for the protective devices before restarting.

9.7 Decommissioning

When the set is finally taken out of service, drain the oil and the cooling water and dispose of them properly.

10 Spare parts

Keep a stock of spare parts for the wearing parts listed below. Order spare parts from the manufacturer and quote the order number and the part number.

Part	Recommended stock
Journal bearing pads, turbine	one set
Thrust bearing pads	one set
Labyrinth seal strips	one set
Oil filter elements	four pieces
Servomotor seal kit	one set

10.1 Ordering spare parts

Quote the order number, the part number from the spare parts list and the required quantity.

10.2 Storage of spare parts

Store spare parts dry and in their original packing. Protect bearing pads against corrosion.

10.3 Spare parts for the oil system

Keep filter elements, pump seals and the oil mist separator element in stock.

10.4 Spare parts for the governor

Keep a servomotor seal kit and a spare speed sensor in stock.

10.5 Spare parts for the generator

Spare parts for the generator are listed in the generator documentation.

Part B Drawings and diagrams

Arrangement drawing, plan view

Drawing sheet: arrangement drawing, plan view. The drawing shows the as-delivered state of the turbo-generator set for order 7.318.042.

Arrangement drawing, side view

Drawing sheet: arrangement drawing, side view. The drawing shows the as-delivered state of the turbo-generator set for order 7.318.042.

Foundation load plan

Drawing sheet: foundation load plan. The drawing shows the as-delivered state of the turbo-generator set for order 7.318.042.

Oil system diagram

Drawing sheet: oil system diagram. The drawing shows the as-delivered state of the turbo-generator set for order 7.318.042.

Oil tank and oil mist separator

Drawing sheet: oil tank and oil mist separator. The drawing shows the as-delivered state of the turbo-generator set for order 7.318.042.

Steam and drain diagram

Drawing sheet: steam and drain diagram. The drawing shows the as-delivered state of the turbo-generator set for order 7.318.042.

Gland steam diagram

Drawing sheet: gland steam diagram. The drawing shows the as-delivered state of the turbo-generator set for order 7.318.042.

Cooling water diagram

Drawing sheet: cooling water diagram. The drawing shows the as-delivered state of the turbo-generator set for order 7.318.042.

Instrument list, turbine

Drawing sheet: instrument list, turbine. The drawing shows the as-delivered state of the turbo-generator set for order 7.318.042.

Instrument list, gearbox and generator

Drawing sheet: instrument list, gearbox and generator. The drawing shows the as-delivered state of the turbo-generator set for order 7.318.042.

Start-up curve, cold condition

Drawing sheet: start-up curve, cold condition. The drawing shows the as-delivered state of the turbo-generator set for order 7.318.042.

Start-up curve, warm condition

Drawing sheet: start-up curve, warm condition. The drawing shows the as-delivered state of the turbo-generator set for order 7.318.042.

Maintenance schedule

Drawing sheet: maintenance schedule. The drawing shows the as-delivered state of the turbo-generator set for order 7.318.042.

Lubrication schedule

Drawing sheet: lubrication schedule. The drawing shows the as-delivered state of the turbo-generator set for order 7.318.042.

Rating plate data

Drawing sheet: rating plate data. The drawing shows the as-delivered state of the turbo-generator set for order 7.318.042.

Site Installation Requirements

for Steam Turbines

Turbo-generator set CB 24 GT4

Order 7.318.042

This specification describes the requirements that the plant owner must meet in the area surrounding the turbine: foundation, alignment, piping, oil system, electrical installation, insulation and monitoring.

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1 General

The turbine supplier delivers the turbine, the gearbox, the generator and the oil system on a common base frame. The plant owner provides the foundation, the connecting piping, the cooling water supply and the electrical connections.

All work must follow the drawings listed in the order documentation. Deviations require written approval by the turbine supplier.

Warning: the installation area must be kept free of flammable material. Hot surfaces of the turbine casing and steam pipes reach more than 400 degrees Celsius during operation.

2 Space requirements

Keep a free area of at least 1.5 m around the base frame for operation and maintenance. Above the turbine, provide a crane hook height that allows the upper casing half and the rotor to be lifted out.

Component	Lifting weight (t)	Clear height above floor (m)
Upper casing half	4.2	5.5
Turbine rotor	2.8	5.0
Generator rotor	6.5	6.0

2.1 Access and ventilation

Provide a transport route for the heaviest single part and a laydown area next to the turbine during inspections. The turbine hall must be ventilated so that the ambient temperature near the oil system stays below 45 degrees Celsius.

Floor openings for piping and cables must be closed with covers that can carry the loads of maintenance work.

3 Foundation design

The foundation must carry the static weight of the turbo-generator set and the dynamic loads during operation, including short-circuit torque of the generator. Design the foundation as a reinforced concrete block or a table foundation on elastic supports.

The natural frequencies of the foundation must stay at least 20 percent away from the running speeds of the turbine and the generator. Grout the base frame with non-shrinking grout after final alignment.

4 Alignment of turbine and driven machine

The turbine, the gearbox and the generator are pre-aligned on the base frame at the factory. After the base frame is set on the foundation, check the alignment of the turbine to the driven machine at every coupling and correct it with shims.

Record the alignment values in the installation report. Check the alignment again after grouting and after the connecting pipes are fitted, because pipe forces can change it.

5 Piping

Connect all pipes free of stress. Pipe forces and moments at the turbine flanges must not exceed the values in the piping drawing.

5.1 Live steam pipe

Install a steam strainer and a quick-closing valve upstream of the turbine. Blow out the live steam pipe before it is connected to the turbine until no particles are found on the target plates.

5.2 Exhaust steam pipe

The exhaust steam pipe leads to the process steam header. Provide an expansion joint and fixed points so that thermal growth does not load the turbine casing.

Install a safety valve in the exhaust steam line that protects the turbine casing against excessive back pressure.

5.3 Drain pipes

Casing drains and valve drains must slope continuously to the drain collector. Do not combine drains of different pressure levels in one pipe without check valves.

Open all drains before start-up so that condensate cannot reach the turbine blading.

5.4 Gland steam pipe

Leak-off steam from the shaft glands is led to a gland steam condenser or to atmosphere at a safe location. The pipe must not rise after leaving the turbine.

5.5 Cooling water

The oil cooler and the generator air cooler need cooling water at the flow rate and temperature given in the data sheet. Provide isolating valves and thermometers at the inlet and outlet of each cooler.

Consumer	Flow (m3/h)	Inlet temperature max. (C)
Oil cooler	12	32
Generator air cooler	38	32

5.6 Blow-off line and start-up against back pressure

During start-up the exhaust steam is blown off to atmosphere until the turbine is synchronized and the process header can take the steam. The required arrangement depends on the back pressure of the process header.

Back pressure lower than 6 bar (g)

Start up against the blow-off valve with the header valve closed. Open the header valve slowly after synchronization and close the blow-off valve at the same rate.

Back pressure higher than 6 bar (g) and lower than 11 bar (g)

Warm up the exhaust pipe through the bypass before start-up. Hold the back pressure with the blow-off control valve until the header pressure is reached.

Back pressure higher than 11 bar (g)

Start up only with the back-pressure controller in automatic mode and with the bypass open. Increase load in steps and watch the axial position of the rotor.

6 Oil system

The oil system supplies lube oil to the bearings of turbine, gearbox and generator and control oil to the governor. Fill the system only with the oil grade given in the lubrication schedule.

Flush the oil system before commissioning with the bearings bypassed until the filter stays clean for four hours.

6.1 Oil tank and oil mist

Vent the oil tank to a safe area. The oil mist separator on the tank vent must be connected with a rising pipe and must not be blocked.

Maintenance of the oil mist separator is described in the operating manual.

7 Electrical installation

Connect the generator, the auxiliary motors and the control cabinet according to the circuit diagrams. Earth the base frame at two opposite corners.

Run signal cables separately from power cables and use shielded cables for speed and vibration sensors.

7.1 Emergency power

The emergency oil pump must be supplied from a battery or an emergency power supply so that the bearings are lubricated during coast-down after a power failure.

8 Insulation

Insulate the turbine casing, the valves and the steam pipes after the pressure test. Leave the flanges of the casing joint and the instrument connections accessible.

Insulation must be protected against oil. Oil-soaked insulation is a fire hazard and must be replaced.

9 Control and monitoring

The plant control system receives the turbine signals for speed, bearing temperatures, vibration, lube oil pressure and axial position. Alarm and trip values are listed in the instrument list.

9.1 Signal exchange

Hard-wired signals are required for the trip commands between the turbine protection and the generator protection. Status and measured values may be exchanged over a fieldbus.

10 Checklist before commissioning

- Foundation grouted and cured.
- Alignment checked after grouting and after piping.
- Live steam pipe blown out and strainer fitted.
- Oil system flushed and filled.
- Cooling water available at the required flow.
- Electrical connections tested.

10.1 Handover

The installation is handed over to the commissioning team when every item of the checklist is signed by the plant owner and the supplier.

Flexible pin type coupling

series KBN 21011

Operating manual KBN 31016, Edition g

Model series FLEXO-N

Role	Name	Date	Signature
Author	Dipl.-Ing. M. Albrecht	12.03.2019	sgd. M. Albrecht
Approved	Dr.-Ing. S. Voigt	18.03.2019	sgd. S. Voigt

This manual applies to FLEXO-N flexible pin type couplings supplied as part of order 7.318.042. Keep it available at the installation site for the whole service life of the coupling.

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1. General information

The FLEXO-N coupling connects two shafts and transmits torque through steel pins fitted with elastic buffers. The buffers damp torsional shocks and compensate small shaft misalignment between the driving and the driven machine.

Warning: carry out work on the coupling only while the drive train is stopped and locked out against unintended start-up. Observe the plant safety instructions and the documentation of the connected machines.

Caution: use only original buffers and pins. Substitute parts change the torsional stiffness of the coupling and can cause damage to the connected machines.

The manufacturer accepts no liability for damage caused by improper assembly, operation outside the rated data, or modifications that were not approved in writing.

2. Features

The coupling consists of a pin half and a buffer half. The pin half carries hardened steel pins; the buffer half carries bores that receive the pins with their elastic buffers. Both halves are fitted to the shaft ends with a cylindrical or tapered bore and a feather key.

- Fail-safe design: if the buffers wear out, the pins still transmit torque.
- Buffers can be replaced without moving the connected machines.
- Suitable for both directions of rotation.
- Permissible operating temperature of the buffers from minus 30 to plus 80 degrees Celsius.

Size	Rated torque (Nm)	Max. speed (1/min)	Bore max. (mm)
125	900	4600	55
160	2100	3600	70
200	4200	2900	90
250	8500	2300	110

2.1 Rated data and selection

The coupling size is selected from the rated torque of the driven machine, multiplied by the service factor of the application. Short-term peak torque during start-up may reach three times the rated torque.

The supplied size for order 7.318.042 is size 200 with a service factor of 1.75 for a steam turbine driving a generator through a gearbox.

Balancing: coupling halves are balanced individually. Mark the relative position of the halves before dismantling so that they can be refitted in the same position.

3. Function

Torque is transmitted from the pin half to the buffer half through the elastic buffers. Under load the buffers deform slightly, which damps torque peaks and allows small radial, angular and axial misalignment.

Before commissioning, inspect the coupling halves, pins and buffers for damage caused by transport or storage. Cracked pins, torn buffers or deformed bores are not acceptable.

If the coupling shows any damage before commissioning, it may not be put into operation. Replace the damaged parts first and record the inspection result in the commissioning report.

A damaged coupling that is put into operation can fail suddenly and endanger persons and the connected machines.

4. Assembly

4.1 Mounting the coupling halves

Clean the shaft ends and the bores. Heat the coupling halves evenly to about 100 degrees Celsius and push them onto the shafts until they are flush with the shaft ends. Do not hammer the halves onto the shafts.

Secure the halves axially with the set screws. Tighten the set screws to the torque given in the assembly drawing.

Move the machines together until the distance between the halves matches the value given in Table 3.

Size	Distance S (mm)	Tolerance (mm)
125	4	+1
160	5	+1
200	5	+1.5
250	6	+1.5

4.2 Aligning

Correct alignment is essential for a long service life of the buffers and the shaft bearings. Align the machines at operating temperature where possible, or allow for the thermal growth given by the machine manufacturer.

4.2.1 Aligning by means of a straight edge and feeler gauge

Place a straight edge across both coupling halves at four points offset by 90 degrees and measure the radial offset with a feeler gauge. Measure the gap between the faces at the same four points to determine the angular offset.

This method is sufficient for sizes up to 160. For larger sizes use a dial gauge.

Permissible shaft misalignment and tolerances

4.2.2 Aligning by means of a dial gauge

Fix the dial gauge to one coupling half and turn both shafts together. Read the radial value on the outer diameter and the axial value on the face of the other half at four positions.

Shaft misalignment compensation of the coupling is limited. The permissible values in Table 4 depend on the nominal size and the speed. Kr is the permissible radial misalignment and Kw the permissible angular misalignment. Stay within the tolerance to protect the buffers.

Nominal size	Speed (1/min)	Kr radial (mm)	Kw angular (mm)
10	1500	0,3	0,7
12	1500	0,3	0,8
16	1500	0,4	0,9
20	1500	0,4	1,0

Table 4: permissible shaft misalignment. For higher speeds reduce the values in proportion to the speed.

5. Commissioning

Before the first start check that the set screws are tightened, the guard is fitted and the alignment values are recorded.

Turn the drive train by hand if possible. The coupling must turn freely without contact between the halves.

Start the drive train and observe the running behaviour. Unusual noise or vibration indicates misalignment or a damaged buffer; stop the machine and check the coupling.

After about 24 operating hours check the alignment again at operating temperature and correct it if necessary.

6. Maintenance

The coupling needs no lubrication. Inspect the buffers for wear at the intervals of the maintenance schedule of the plant, at the latest every 4000 operating hours.

Measure the torsional backlash between the halves. If the backlash exceeds the wear limit in the assembly drawing, replace all buffers of the coupling at the same time.

Check the alignment during each inspection. A change in alignment often indicates a foundation or bearing problem; find the cause before correcting the alignment.

Caution: damage to buffers by oil or solvents shortens their service life. Keep the coupling clean and dry.

7. Dismantling

Secure the drive train against start-up. Remove the guard and mark the position of the coupling halves.

Move the driven machine back until the pins leave the buffer half. Pull the coupling halves off the shafts with a suitable puller; heat them if necessary.

Replace worn buffers and damaged pins with original parts before reassembly.

8. Declaration of the manufacturer

The FLEXO-N coupling is intended to be incorporated into a machine. It must not be put into service until the machine into which it is incorporated has been declared in conformity with the applicable machinery regulations.

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Folder 2 Accessory documentation

This folder contains the documentation of the lube oil purifier, the turning gear and the steam trap station supplied with the turbo-generator set.

Lube oil purifier: overview

The lube oil purifier is a centrifugal separator that removes water and particles from the turbine lube oil in a side stream. It runs continuously while the oil system is in operation.

Warning: the separator bowl rotates at high speed. Never open the purifier before the bowl has come to a complete stop.

Lube oil purifier: start-up

Before start-up check the gearbox oil level of the purifier and the direction of rotation. Caution: running the bowl in the wrong direction causes damage to the drive.

Lube oil purifier: bowl cleaning

Clean the separator bowl every 500 operating hours, or earlier when the discharge alarm occurs. Tighten the bowl lock ring to the torque stamped on the ring.

Warning: an incorrectly tightened lock ring can loosen during operation and cause serious damage. Caution: use only the special tools supplied with the purifier.

Lube oil purifier: alarms and trip

The purifier raises an alarm for high vibration, low feed pressure and failed sludge discharge. High vibration trips the purifier motor.

Fault finding: after a trip, check the bowl for sludge build-up and imbalance before restarting. Warning: do not restart after a vibration trip until the cause is found.

Lube oil purifier: maintenance schedule

Maintenance intervals: bowl cleaning every 500 hours, inspection of the bowl seals every 2000 hours, overhaul of the drive every 8000 hours. Record all maintenance in the purifier log.

Lube oil purifier: lubrication

Lubrication of the purifier gearbox: change the gearbox oil every 2000 hours. Check the oil level weekly. Caution: overfilling the gearbox causes overheating of the drive.

Lube oil purifier: troubleshooting

Trouble: water in the cleaned oil. Cause: wrong gravity disc or feed temperature too low. Remedy: fit the correct gravity disc and raise the feed temperature to the setpoint.

Trouble: sludge discharge fails. Cause: blocked operating water line. Remedy: clean the operating water strainer. A repeated discharge fault raises an alarm.

Lube oil purifier: setpoints

Feed temperature setpoint, feed pressure alarm limit and vibration trip limit are listed in the purifier data sheet. Changing a setpoint requires the approval of the plant engineer.

Lube oil purifier: spare parts

Keep a set of bowl seals, a gravity disc set and a spare operating water valve in stock.

Turning gear: overview

The turning gear rotates the turbine rotor slowly after shutdown so that it cools evenly and does not bow. It engages automatically at standstill and disengages when the turbine starts.

Turning gear: operation

Start the turning gear only when the lube oil pressure is available. Warning: the turning gear must not be engaged while the rotor is turning above turning speed; damage to the gear teeth results.

Turning gear: alarms and faults

The turning gear raises an alarm when it fails to engage and a fault when the motor current is too high. A high motor current indicates rubbing in the turbine; stop the turning gear and investigate. Caution: do not force the rotor with the hand lever.

Turning gear: maintenance

Maintenance: inspect the engagement mechanism every 4000 hours and check the gear teeth for wear.
Lubrication of the gear follows the lubrication schedule.

Turning gear: commissioning checks

During commissioning check engagement and disengagement at standstill, the motor direction of rotation and the interlock with the lube oil pressure. Alignment of the turning gear motor is set at the factory.

Turning gear: torque limiter

A torque limiter protects the turning gear against overload. Check the torque setting at each major inspection; damage to the limiter plates requires replacement.

Steam trap station: overview

The steam trap station drains condensate from the live steam line and the casing drains. Each trap has an isolating valve and a test valve.

Steam trap station: testing

Test each steam trap during rounds by opening the test valve. A trap that blows live steam continuously is faulty and must be replaced.

Steam trap station: maintenance

Maintenance: clean the trap strainers every 2000 hours. Caution: isolate and depressurize the trap before opening it. Warning: hot condensate causes burns.

Steam trap station: troubleshooting

Trouble: water hammer in the drain line. Cause: a blocked steam trap. Remedy: clean or replace the trap; check the alignment of the drain line slope. A malfunction of the drain valves can cause damage to the turbine blading.

Steam trap station: alarms

A high-level alarm in the drain pot indicates a failed trap. After the alarm, check the trap and the drain valve before restarting the turbine; a trip on drain pot level protects the blading.

Steam trap station: spare parts

Keep a spare trap of each size and a set of strainer screens in stock.

Vibration monitoring rack: overview

The vibration monitoring rack measures shaft vibration and bearing housing vibration. It raises an alarm at the first limit and trips the turbine at the second limit.

Warning: never bridge a trip channel during operation. A fault in a channel is shown on the rack and raises a fault alarm.

Vibration monitoring rack: maintenance and calibration

Maintenance: check the sensor gaps at each major inspection and calibrate the channels. Caution: a wrong setpoint after calibration causes a false trip or no trip at all; damage to the bearings can follow.

Vibration monitoring rack: commissioning

During commissioning verify every alarm and trip setpoint by injecting a test signal. Check the alignment of each probe bracket and tighten it to the specified torque.

Emergency oil pump battery charger

The battery charger keeps the emergency oil pump battery ready. A charger fault raises an alarm. Warning: a discharged battery means no bearing lubrication during coast-down after a trip; damage to the bearings results. Maintenance: test the battery capacity every year.

Emergency oil pump battery: troubleshooting

Trouble: the emergency oil pump does not start on test. Cause: battery fault or contactor malfunction. Remedy: check the battery voltage and the contactor. Danger: battery rooms contain explosive gas; ventilate before work.

Gland steam condenser: operation and alarms

The gland steam condenser keeps the gland leak-off under slight vacuum. A high-level alarm or a fan fault indicates a malfunction. Caution: without vacuum, steam escapes at the shaft seals and can cause damage to bearings through water in the oil.

Gland steam condenser: maintenance

Maintenance: clean the condenser tubes and check the fan every 4000 hours. Lubrication of the fan bearings follows the lubrication schedule. Check the fan alignment and belt torque.

Overspeed test device: operation

The overspeed test device lets the operator test the mechanical overspeed trip below rated speed. Warning: the turbine must be at no load. A failed test is a fault that must be eliminated before commissioning or restarting.

Overspeed test device: setpoints and damage

The trip setpoint of the test device is set at the factory. Caution: changing the setpoint without approval is not permitted. Damage to the test piston causes a malfunction of the trip; replace the piston during maintenance.

Accessory maintenance summary

Maintenance, lubrication and alarm test intervals of all accessories are summarised in the maintenance schedule. Warning: skipping an alarm or trip test can hide a fault until damage occurs. Torque values and setpoints are listed in the accessory data sheets.